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Sales Performance Analysis & Reporting - SPAR Project Lifecycle Documentation

This document contains a full space export of the SPAR documentation environment. It captures the end-to-end operational roadmap, data asset lineage records, governance standards, and delivery post-mortems utilized during the project deployment cycle

Export Scope : 5 Core Workspace Pages

Space Key : SPAR

Author : Snigdha Nath

Role : IT Project Manager & Data Analyst

Integrated Tools : Atlassian Jira, Confluence, Power Query, Excel, Powerpoint



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Project Charter: Sales Performance Analysis & Reporting (SPAR)

1. Project Background & Objective

- **Background:** Sales leadership currently lacks a centralized, automated system to analyze regional sales performance for FY2024. Transaction data is trapped in manual, disconnected Excel sheets, leading to delayed insights, human error, and inconsistent regional metric reporting.
- **Objective:** To design, build, and deploy a standardized Excel executive dashboard that serves as a single source of truth (SSOT). The goal is to reduce reporting latency by automating data ingestion and providing interactive, drill-down visibility into 5 core sales KPIs.

2. Project Scope

- Ingestion, cleaning, and structural transformation of raw FY2024 transactional datasets.
- Development of a relational data model using data architecture best practices.
- Creation of interactive visualization assets including KPI tracking cards, trend lines, and dynamic regional slicers.
- Complete end-to-end documentation of data governance and transformation logic.

3. Tech Stack & Collaboration Tools

Tool	Project Purpose / Application
Microsoft Excel	Primary repository for raw transactional data inputs and end-user spreadsheet reporting.
Power Query	Extract, Transform, Load (ETL) engine used to clean data, fix mismatches, and shape the data schema.
Atlassian Jira	Agile project management platform utilized for backlog grooming, sprint planning, and velocity tracking.
Atlassian Confluence	Product documentation hub hosting the project charter, data dictionary, ETL logs, and guides.

4. Agile Timeline (3-Sprint Lifecycle)

- **Sprint 1 — Data Architecture & ETL:** Data connection, ingestion, column profiling, handling missing values, and establishing the schema. [Status: COMPLETED]
- **Sprint 2 — Front End Development:** Building pivot tables, dashboard layout wireframing, and KPI metric card configuration. [Status: COMPLETED]
- **Sprint 3 — UAT, Bug Tracking & Launch:** Execution of stakeholder User Acceptance Testing (UAT), fixing critical bugs, and publishing final project documentation. [Status: IN PROGRESS]

5. Project Stakeholders & Roles

- **Product Manager / Scrum Master (Snigdha Nath):** Responsible for scope definition, timeline management, Jira backlog prioritization, roadblock removal, and stakeholder delivery alignment.
- **Data Analyst (Snigdha Nath):** Responsible for executing the technical data pipeline, writing transformation steps in Power Query, modeling the tables, and designing the frontend interactive dashboard interfaces.

6. Success Criteria

- **Data Quality Alignment:** 100% elimination of null values and formatting mismatches across core fields before visualization.
- **Operational Efficiency:** Reducing the time required to generate regional sales summaries from hours of manual work down to a single-click dashboard refresh.
- **Adoption Rate:** Successful delivery of user documentation enabling non-technical regional sales managers to extract weekly insights independently.

Data Dictionary: SPAR Sales Dataset

Overview

This document serves as the data governance single source of truth for the **SPAR** project. It details the structural schema, data types, business logic, and formatting standards applied to the finalized, clean transactional sales dataset. All raw source anomalies have been resolved through the ETL process prior to schema optimization.

Core Sales Transaction Schema

Column Name	Data Type	Business Description	Transformation/Calculated Logic	Example Value
Order_ID	Text (String)	The unique alphanumeric identifier assigned to each individual sales transaction.	Unique key; leading/trailing spaces trimmed.	ORD-6633
Order_Date	Date	The calendar date when the transaction occurred.	Standardized to DD/MM/YYYY format.	29/03/2024
Month Name	Text (String)	The text name of the month derived from the transaction date.	Extracted from Order_Date via Power Query for time-series filtering.	March

Quarter	Text (String)	The business calendar quarter derived from the transaction date.	Calculated from Order_Date (Q1 , Q2 , Q3 , Q4) for executive reporting.	Q1
Product	Text (String)	The specific name of the item sold to the customer.	Grouped and standardized to eliminate typos.	Laptop Pro
Region	Text (String)	The geographic business territory where the sale was executed.	Cleansed to match official corporate operating zones.	South
Sales_Rep	Text (String)	The full name of the internal sales representative responsible for the account.	Standardized to proper text casing (First Last).	Carol White
Quantity	Number (Integer)	The total number of units purchased within the specific order line item.	Validated integer; restricted to values greater than 0.	7

Unit_Price	Currency	The individual unit cost of the product at the time of sale.	Formatted as standard regional currency (\$).	\$324
Total_Amount	Currency	The gross revenue generated by the specific transaction line item.	Calculated field: $\$ \text{\text{Quantity}} \times \text{\text{Unit_Price}} \$$	\$2,268
Status	Text (String)	The active fulfillment operational status of the order.	Categorized into uniform states (e.g., Delivered, Pending).	Delivered

Data Integrity & Governance Constraints

- **Primary Key:** Order_ID must contain unique values and cannot contain null or empty strings.
- **Mathematical Alignment:** Total_Amount must always strictly equal the product of Quantity and Unit_Price. Any discrepancies indicate upstream ingestion errors.
- **Temporal Hierarchy:** Month Name and Quarter values must always logically match the corresponding Order_Date.

Data Quality & Cleaning Log (ETL Lineage)

Overview

Prior to data modeling and visualization, the raw FY2024 transactional sales dataset underwent an extensive Extract, Transform, Load (ETL) lifecycle within **Power Query**. The primary objective was to enforce data integrity, eliminate structural anomalies, and establish a clean, consistent dataset. This log archives the step-by-step data transformations and their volumetric impact.

Power Query Transformation Audit Trail

Step	Applied Transformation (Action)	Technical & Business Justification (Reason)	Volumetric Impact
0	Raw Ingestion	Initial connection to the unstructured historical Excel workbook.	Ingested: 305 Rows
1	Removed Blank Rows	Extracted 4 completely unpopulated record rows manually from the source grid.	\$305 \rightarrow 301\$ Rows
2	Removed Duplicate Records	Isolated and deleted duplicate <code>Order_ID</code> entries representing duplicate transactions.	\$301 \rightarrow 270\$ Rows
3	Standardized Text Columns	Trimmed trailing whitespaces and converted text	<i>No row loss</i>

		fields (Product , Region , Sales_Rep) to proper case to resolve inconsistent typing.	
4	Normalized Mixed Date Formats	Resolved 4 conflicting date formats found within the raw Order_Date field. Corrupted strings unable to parse were removed.	<i>Minor row loss</i>
5	Filtered Null Date Rows	Purged remaining records missing an Order_Date stamp, as they were unusable for downstream time-series analysis.	\$270 \rightarrow 259\$ Rows
6	Derived Temporal Columns	Extracted calculated dimensions (Month Name and Quarter) directly from the clean Order_Date column to	<i>No row loss</i>

		support analytical pivot routing.	
—	Finalized Data Output	Successful transformation completion. Clean dataset loaded into the reporting model.	Final Output: 240 Rows

Key Engineering Decisions & Data Integrity Rules

- **Handling Null Dates:** Records without an absolute transaction date were classified as operational waste. Because a timeline anchor is mandatory for rolling sales metrics, dropping these rows was chosen over using placeholder fallback dates to protect the accuracy of time-series trends.
- **De-duplication Policy:** `Order_ID` must represent an atomic, unique sale. Duplicate records were evaluated as system data logging glitches and stripped to prevent artificial revenue inflation.
- **Case Normalization:** Ensuring uniform syntax across dimensions like `Region` avoids fragmented visualization filters (e.g., preventing "South" and "south" from displaying as distinct reporting groups).



Dashboard User Guide: Executive Sales Performance View

Overview

The **Executive Sales Dashboard** provides a centralized enterprise interface for analyzing macro sales performance, monitoring operational order fulfillment states, and auditing sales representative performance metrics. This documentation serves as an onboarding guide for non-technical business stakeholders, outlining the operational procedures required to navigate interactive visualizations, execute multi-dimensional cross-filtering, and trigger manual data refresh loops securely.

1. Core Summary Metrics (KPI Cards)

Positioned horizontally across the primary header row of the dashboard layout, these baseline metrics calculate and update dynamically based on active filter criteria:

- **Total Revenue (\$366 , 751)**: Represents the cumulative gross revenue generated across all successful transactional entries within the system database.
- **Total Orders (1 , 244)**: Tracks the aggregate count of unique transactional order identifiers processed during the reporting period.
- **Top Product (Usb Hub)**: Identifies the highest-performing inventory line item based strictly on total transactional sales volume.
- **Return Rate (26%)**: Evaluates overall operational health by calculating the exact proportion of order rows marked with a fulfillment status of "Returned" against total processed orders.

2. Dynamic Filtering & Segmentation (Interactive Slicers)

Multi-dimensional data slicing is controlled via the dedicated filtering panel configured along the right-hand margin of the dashboard canvas:

- **Region Slicer**: Segregates global performance metrics into specific geographic operating territories (**East, North, South, West**). Selection automatically adjusts peripheral visualizations, including the *Revenue by Region* pie chart and the *Sales Rep Performance* bar matrix.
- **Quarter Slicer**: Isolates temporal performance windows into specific fiscal quarters (**Q1, Q2, Q3, Q4**) to support seasonal trend auditing and quarterly business reviews.
- **Product Slicer**: Filters the entire visualization canvas down to individual product assets (e.g., *Headset Pro, Keyboard Mechanical, Usb Hub*) to isolate individual product performance trends.

 **System Operational Standard (Multi-Select):** To apply compound filtering across multiple separate dimensions simultaneously, hold down the **Ctrl** key (or **Cmd** on macOS) while making selections within any interactive slicer panel.

3. Filter Reset Procedures

When analytical metrics appear heavily constrained, or when reverting the dashboard view to its global baseline is necessary, standard reset actions include:

1. Hover the cursor over the top-right header margin of the targeted slicer component (**Region, Product, or Quarter**).
2. Locate and select the contextual **Clear Filter** icon (represented visually by a funnel graphic overlaid with an "X").
3. Alternatively, clicking an active, highlighted filter dimension a second time will natively deselect that specific value and restore the global dataset display.

4. Data Refresh Protocol (Ingesting New Transaction Rows)

To successfully synchronize the interactive dashboard layout when new row entries are appended to the historical Excel data source, the following structured pipeline must be executed:

1. **Source Data Preparation:** Open the localized source spreadsheet infrastructure, append the new transactional entries, and manually verify that all updated records strictly conform to the structural data types and naming conventions defined on **Page 2 (Data Dictionary)**. Save and close the file.
2. **Interface Initialization:** Access the primary reporting application interface and locate the application control ribbon.
3. **Command Execution:** Navigate to either the **Data** or **Home** tab selection and execute the **Refresh All** (or **Refresh**) command.
4. **Automated Processing Lifecycle:** The background **Power Query** processing engine will automatically initialize, executing the sequential transformation steps explicitly logged on **Page 3 (Data Quality & Cleaning Log)**. This updates the frontend visual data models and KPI summaries instantaneously without manual recalculation.

Project Retrospective: SPAR Delivery Lifecycle

Overview

This retrospective serves as an honest post-mortem analysis of the **SPAR (Sales Performance Analysis & Reporting)** project lifecycle. It evaluates the technical execution of the data pipeline, the project management choices made within Jira, and strategic improvements identified for future product delivery cycles.

1. What Went Well (Successes)

- **Highly Efficient ETL Pipeline:** Enforcing automation rules within **Power Query** allowed the data preparation pipeline to run seamlessly. Complex cleanup tasks—such as stripping unpopulated rows and handling duplicate entries—were executed systematically without damaging the core relational model schema.
- **Interactive Dynamic Delivery:** The frontend dashboard interface successfully achieved full cross-filtering capabilities. Business stakeholders can toggle multi-dimensional slicers smoothly, providing an intuitive, interactive user experience that simplifies executive decision-making.
- **End-to-End Delivery Realism:** Managing the documentation timeline concurrently with active development sprints established a complete, audit-ready product environment from kickoff to closure.

2. What Was Challenging (Roadblocks & Constraints)

- **Structural Source Vulnerabilities:** The raw data source contained four conflicting date formatting structures. Parsing these mismatched strings caused initial loading errors within the query engine and required manual structural profiling to resolve.
- **Strategic Data Attrition:** Resolving corruption in data fields resulted in an approximate **15% total data loss** (\$305 \rightarrow 240\$ rows) across the transactional database.

3. Lessons Learned (Key Takeaways)

- Memorializing Power Query steps and metric logic inside Confluence ensures the reporting framework remains maintainable and accessible to future technical teams.
- Story points were not assigned prior to sprint initiation, resulting in empty velocity and burndown reports. In future sprints, story point estimation will be completed during sprint planning before sprint start .

4. Future Actions (What to Do Differently Next Time)

- In future project lifecycles, data validation rules (such as input masks, strict column types, and data entry constraints) must be enforced at the initial ingestion or data collection point before the dataset enters the staging environment.
- Story point estimation will be completed during sprint planning before sprint start .